

Madhusudan Sharma

PhD

GEO3BCN-CSIC

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Degree	Department	University	Year
PhD (ongoing)	GEO3BCN-CSIC	University of Barcelona	2024-present
Master of Science	Earth Science	IIT Bombay	2021-23
Bachelor of Science	Physics	BHU, Varanasi	2017-20

RESEARCH EXPERIENCE

• PhD

(GEO3BCN-CSIC, Geodynamics lithosphere group | Supervisor: **Dr. Ivone Jiménez munt**) (July 2024- present)

- Title - Geodynamics and Evolution of the Adria Microplate: From the Mantle to the Surface Through Numerical Modeling
- Performed 2D numerical geodynamic modelling of subduction zones, inspired by the Calabria and Hellenic arcs.
- Courses
 - Specific English : Writing Online course offered by CSIC. (05/19/20205 – 06/20/2025)
 - TALLER DE SUPERCOMPUTACIÓN: USO DE LOS RECURSOS HPC DE SGAI-CSIC (2-06-2025 to 5-06-2025)
 - Lithosphere Dynamics and topography a Master's courses in Reservoir Geology and Geophysics (Dr. Daniel Garcia-Castellanos, Dr. Ivone Jiménez-Munt, Dr. Maria Gema Llorens, Prof. Jaume Vergés Masip).
- STAGE (GFZ Potsdam, Section 4.1 Geodynamics Lithosphere | Supervisor: **Prof. Claudio Faccena**) (October 2025- December 2025)
 - Gain experience in 3D subduction modelling setup

• Junior Research Fellow

(Center of Earth Sciences, Geophysical Inversion Lab, IISc | Guide: **Dr. Pawan Bharadwaj**) (Aug 2023- May 2024)

- Title - Time reversal imaging using virtual seismograms
 - [Report Link](#)
 - [Link for time reversal movies](#)
- Contributed to the development of earthquake project into interactive workflow automation software in Julia
- Performed synthetic test for validation of the algorithm
- Performed cluster analysis of latent code using the t-SNE package
- Incorporating surface-reflected depth phases for time reversal imaging
- Scaling earthquake source project for large numbers of earthquakes

• Master's Project

(Indian Institute of Technology, Bombay | Guide: **Dr. Anand Singh**) (Jan 2023- April 2023)

- Title - 2D Inversion of Self-Potential Data for the Ambet Region
 - [Thesis Link](#)
- Investigated potential subsurface water bodies in the Ambet region
- Employed triangular mesh grid for numerical modelling
- Performed compact inversion technique

• Research Intern

(Center of Earth Sciences, Geophysical Inversion Lab, IISc | Guide: **Dr. Pawan Bharadwaj**) (May 2022- July 2023)

- Title - Learning earthquake sources using symmetric autoencoder
 - [Report Link](#)
- Trained symmetric autoencoder with GPU using CUDA
- Performed redatuming and generate virtual seismograms
- Explored similarity between collocated earthquakes using symmetric autoencoder
- Contributed to a comprehensive literature review on various earthquakes, validating findings and enhancing earthquake seismology knowledge

• Summer Intern

(Center of Earth Sciences, Geophysical Inversion Lab, IISc | Guide: **Dr. Pawan Bharadwaj**) (April 2021- July 2021)

- Presented case studies on fluvial, volcano and glacial seismology
 - [Internship Reort](#)

RESEARCH PUBLICATION AND CONFERENCE

- **Madhusudan Sharma**, Ivone Jiménez-Munt, Ana María Negredo Moreno, Ángela María Gómez-García, Michael Pons, Claudio Faccenna, Jaume Vergés, Daniel García-Castellanos, Wentao Zhang and Montserrat Torne: Subduction Dynamics driven by Crustal Heterogeneities in the Subducting Plate (Submitted, *Frontiers in Earth science*)
- Pawan Bharadwaj, **Madhusudan Sharma**, Isha Lohan, Pragna Sahoo; Learning Earthquake Sources Using Symmetric Autoencoders. *Bulletin of the Seismological Society of America* 2025;; 115 (6): 2553–2577

Conference Proceedings

- **M. Sharma**, Ivone Jiménez-Munt, Ana María Negredo Moreno, Ángela María Gómez-García, Michael Pons, Claudio Faccenna, Jaume Vergés, Montserrat Torne, Wentao Zhang, and Daniel García-Castellanos " Slab breakoff induced by crustal-scale heterogeneity "EGU General Assembly 2026 (Oral presentation)
- **M. Sharma**, Ivone Jiménez-Munt , Ana María Negredo Moreno, Ángela María Gómez-García , Jaume Vergés , Montserrat Torne , Wentao Zhang and Daniel García-Castellanos "Trench retreat and slab geometry controlled by crustal heterogeneities of the subducting plate" GEOMOD 2025, Lisbon (Poster)
- **M. Sharma**, Ivone Jiménez-Munt , Ana María Negredo Moreno "Investigating Slab Dynamics through Thermo-Mechanical Modelling Techniques " Geomare speeches 2024, Barcelona (Poster)
- I. Lohan, **M. Sharma**, A. P. Singh, and P. Bharadwaj, "Making Uniform Scattering Seismograms for Earthquake Analysis with Symmetric Autoencoders" in National Workshop on "Earth Sciences in India: Challenges and Emerging Trends (ESICET-2023)", IIT Roorkee
- I. Lohan, **M. Sharma**, A. P. Singh, and P. Bharadwaj, [Earthquake analysis with virtual seismograms:Leveraging symmetric autoencoders for time-reversal imaging](#), in AGU 23, Poster Presentation, 2023.
- I. Lohan, **M. Sharma**, A. P. Singh, and P. Bharadwaj, "Far field earthquake analysis with symmetric autoencoder," in Two Day National seminar on Machine Learning and Deep Learning Applications in Geosciences, Indian Institute of Technology, Mumbai, 2023

COMPUTATIONAL SKILLS & CERTIFICATION

- **Programming Languages:** Python, Julia, MATLAB, Fortran, Shell script
- **Operating System:** Linux, Windows
- **Software:** ASPECT, Specfem 3D, SeiSee, Seismic Unix, Surfer
- **Libraries:** ObsPy, TensorFlow, PyGMT, t-SNE, Seis, Flux, Random, MLUtils, DSP, PlutoPlotly
- **Tools:** L^AT_EX, GitHub, Syngine, Inkscape, Microsoft Office
- **Certification:** Complete Python Bootcamp From Zero to Hero in Python, Geospatial Applications for Disaster Risk Management, RS And GIS Applications, National Anveshika Experimental Skill Test, Classical Electromagnetism-1, Basics Of Quantum Mechanics, Wikipedia interaction workshop, Basics of Special Theory of Relativity, Understanding Design workshop

FIELD & PRACTICAL EXPERIENCE

- **Seismic Refraction Survey** | *Gymkhana Ground, IIT Bombay*
Guide: Dr. Bharath Shekhar, Associate Professor IIT Bombay
 - Acquired 2D Seismic refraction data over an area using 24 channel geophone arrays
- **Electrical Survey** | *Community hall, IIT Bombay*
Guide: Dr. Anand Kumar Singh, Assistant Professor, IIT Bombay
 - Underwent training in resistivity data acquisition both in sounding and profiling using a DC resistivity meter.
- **Geological Field work** | *Mumbai, Maharashtra*
 - Study of Lava flow type, Lava crust, Pipe Vesicles, Spherical vesicles | Elephanta Island, Deccan Traps, Mumbai
- **IISc Open Day** | *Indian Institute of Science, Bengaluru*
 - Title - Quake Songs: Investigating An Earthquake By Hearing Its Song
 - Description - We are constructing a model of an earthquake machine, sonifying the waves it generates. Additionally, we are not only generating sounds from synthetic earthquakes but also with real earthquakes using our USVS
 - Role - Leading this project

AWARDS & ACHIEVEMENTS

- I achieved a percentile rank of 95.5% in the IIT JAM Physics exam and have been successfully admitted to the M.Sc. Applied Geophysics program at IIT Bombay
- Among the top 20% in my M.Sc. batch
- Awarded for qualifying NGPE (National Graduate Physics Exam)